SELECTED WORKS · 2022 – 2026

Shen Linkun

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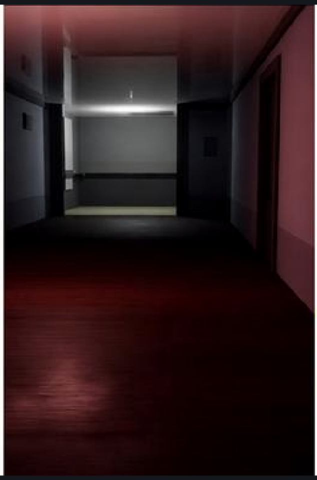
Game designer working at the seam between **systems design**, **technical implementation**, and **AI-assisted content production**. Systems Designer at Tencent TiMi Studio Group · B.S. Games, University of Utah.

Selected Works · Online Edition

All work is original and created by the applicant.
Generative-AI use is disclosed on the final page.

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ABOUT

I design systems — and build the pipelines that make them producible.

I came into games through design and stayed because of the machinery underneath it. Across four solo/independent titles and two years inside a live mobile MOBA, the thread in my work has been the same: **a design is only as good as the production system that can actually ship it a hundred times over.**

That belief pushed me from writing design documents into writing the tools — behaviour trees, blueprint logic, editor scripts, configuration generators — and, more recently, into **AI-assisted content pipelines**: image generation, image-to-3D, retopology, rigging, and the validation gates that keep generated assets from silently breaking a build.

What I want from a graduate programme is the other half of that loop: the theory, the critical vocabulary, and a studio culture where the question isn't only "can this be produced?" but "what is worth producing?"

HOW I WORK

— **Framework first, features second.** Structure the system, then fill it.

— **Instrument everything.** If I can't measure the experience, I'm guessing.

— **Automate the repetition, never the judgement.** AI proposes; a human gate approves.

CAPABILITIES

Design
Systems & economy · level design · combat & boss encounters · environmental / looping narrative

Engineering
Unreal Blueprints · behaviour trees · Python tooling · Flask / SQL · CI test gates

Art & Tech Art
Maya · MotionBuilder · ZBrush · Blender · retopology · skinning · PBR · rig retargeting

AI Production
Image generation · image→3D · ComfyUI local deployment · RAG pipelines · multi-model API middleware

CONTENTS

Unreal Engine · Unity · Maya · Blender · ZBrush · MotionBuilder · Python · Flask · ComfyUI · Behaviour Trees · Git

I · GAME DESIGN — 01

Dream

A first-person psychological horror game about reading a building that refuses to explain itself.

ROLE	ENGINE	DURATION	OUTSOURCED
Solo — design & all Blueprint implementation	Unreal Engine 5	Aug – Dec 2024	Selected art & audio assets only

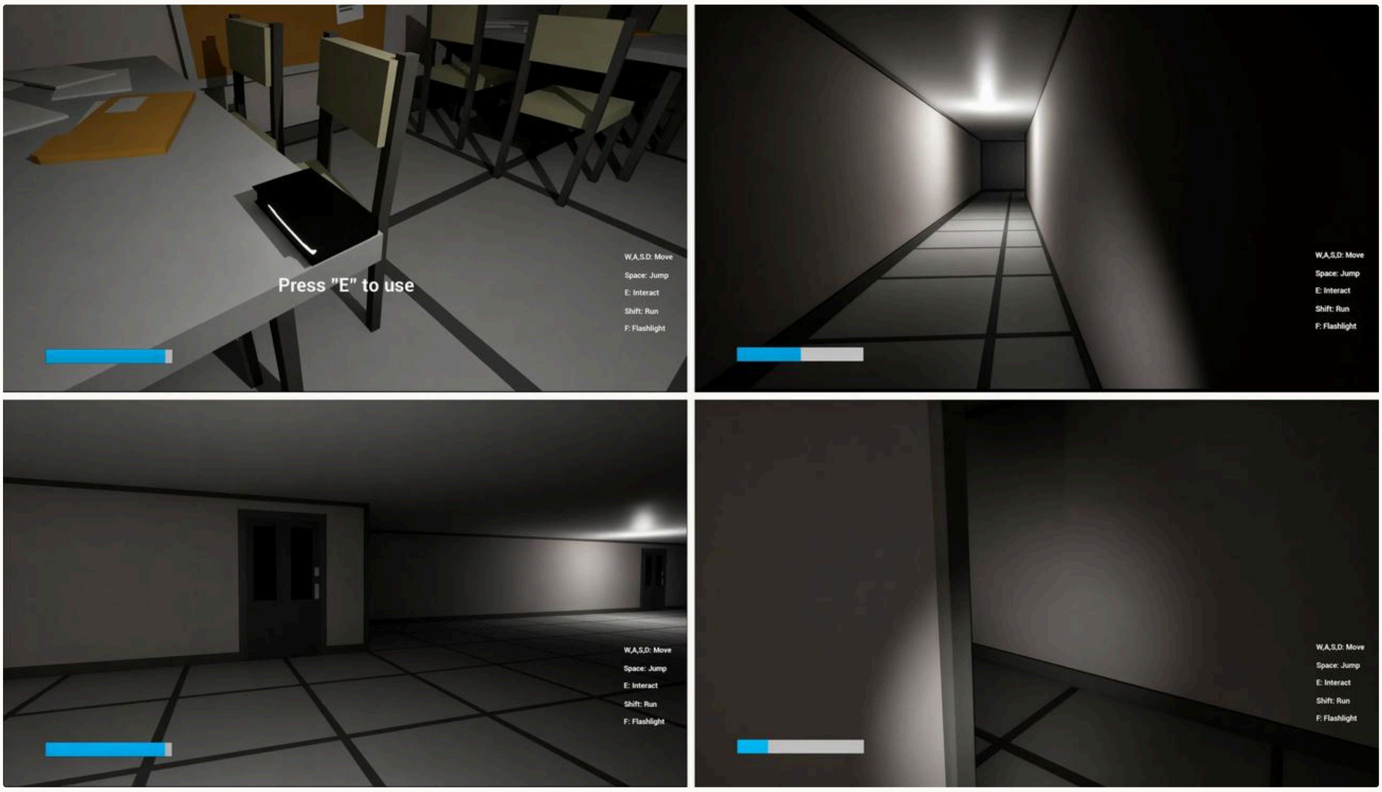
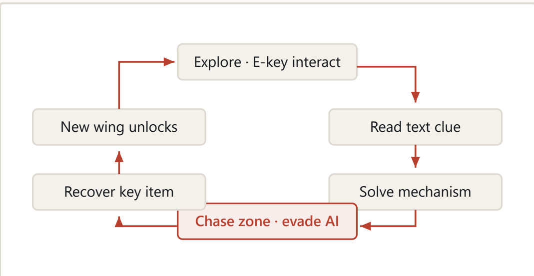


Fig 1–4. Walk, look, press E — the whole verb set. One light at the vanishing point; the flashlight is the only argument against the dark.

THE CORE LOOP



Exploration is the loop; pursuit is a **rare branch off it**. I deliberately kept the chase scarce — constant threat becomes noise and the player stops feeling it, whereas a threat that has fired once and could fire again turns every ordinary corridor into a possible corridor.

Takeaway — mid-project I refactored the Blueprint layer into explicit categories; iteration speed roughly doubled.

DESIGN DECISIONS

— **The threat is rare on purpose.** Pursuit occupies only a small fraction of the running time. Dread does not come from being chased often — it comes from having been chased once and not knowing when it will happen again. Most of the game is the waiting.

— **The AI must be able to lose.** The pursuer disengages once the player exits its detection region. A monster that can never be escaped stops being a threat and becomes a timer.

— **State changes are diegetic.** Triggering a mechanism swaps the score and shifts the practical lights — the player learns the rule of the space without a single line of UI.

— **Resources meter the courage.** A flashlight and a finite stamina bar are the only systems the player manages. Both spend fastest exactly when they are most needed, so caution has a price and panic has a bigger one.

Character presentation state machine

Sole owner of the animation state architecture behind a collectible companion system in a live mobile MOBA.

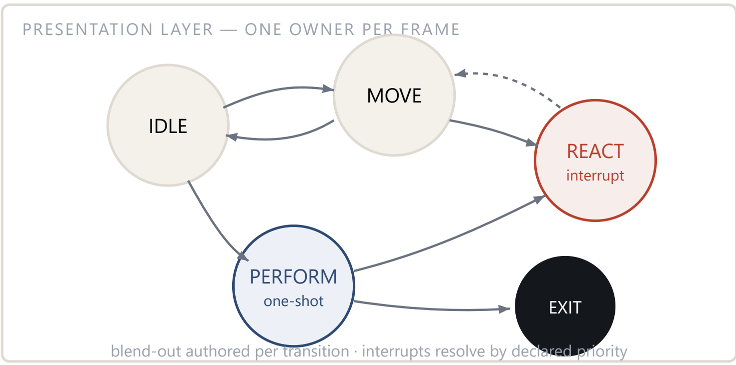
CONTEXT	TITLE	PERIOD	OWNERSHIP
Tencent TiMi Studio Group · Systems Designer	Nation-wide mobile MOBA (live service)	Aug 2025 – present	Sole designer for the state architecture

UNDER NDA No proprietary screenshots, configuration data or engine files are shown. All diagrams in Section II are re-drawn by me at a generic, non-identifying level of abstraction.

THE PROBLEM

Out-of-match companions have to idle, wander, react to the player, respond to taps, and play one-off performance beats — while the player is simultaneously browsing menus, changing outfits and entering queue. The failure mode is not a missing animation; it is a **bad seam**: a companion snapping mid-stride, or freezing because two systems each believed they owned the character that frame.

THE ARCHITECTURE



— **Explicit single ownership.** Exactly one state drives the character each frame; every other system requests rather than sets. This removed the entire class of "two systems fighting" bugs.

— **Transitions are content, not glue.** Each edge has its own authored blend-out, so a walk that is interrupted by a tap decelerates instead of cutting.

— **Interrupts resolve by declared priority.** A reaction can always take the character from idle or move, but never from a one-shot performance mid-beat.

OUTCOME

Out-of-match presentation reads as continuous motion rather than a sequence of clips. Concretely: no more visible pops at the idle↔move seam, and reaction beats can be triggered from any menu context without the animation team hand-authoring a case for each one.

WHAT I ACTUALLY DID

— Enumerated every behaviour the feature promised, then cut the state list to the smallest set that could express all of them.

— Wrote the transition matrix — including the illegal transitions, which is where most of the design lives.

— Specified blend timings against the animation team's clips rather than assuming defaults.

— Sat in review with engineering and animation until the seam list was empty.

DESIGN PRINCIPLE CARRIED FORWARD

A state machine is a contract about who is allowed to lie to the player. Writing down which transitions are forbidden turned out to be far more valuable than enumerating the permitted ones.

State machines · Animation systems · Cross-discipline spec · Live service

Gameplay-mode logic: skills, monster AI, boss phases

Adapting a competitive hero roster into an entertainment battle-royale mode — and rebuilding the enemy AI that populates it.

UNDER NDA Structures below are re-drawn generically; no proprietary logic assets are reproduced.

RE-TUNING SKILLS FOR A DIFFERENT CONTRACT

A competitive MOBA kit is balanced for a 15-minute macro game. Dropped into a fast entertainment mode, the same kit reads as sluggish — long cooldowns and resource costs that exist to create macro decisions instead just create waiting. I worked through the roster adjusting cooldowns, costs and trigger conditions via hero logic graphs and buff triggers so the kits kept their identity while matching the new tempo.

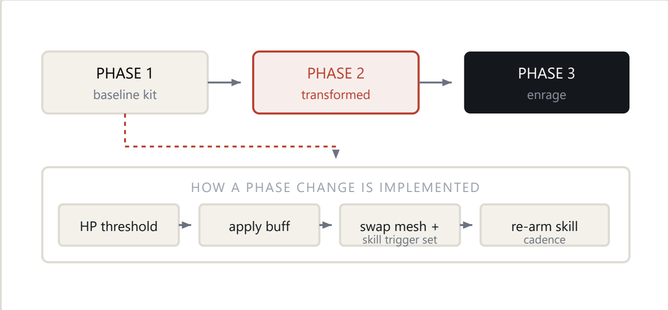
> THE RULE I WORKED TO
>
> Change the **rhythm**, never the **fantasy**. A player who mains a hero should still recognise their hero; what changes is how often they get to be them.

WHERE THE AI ACTUALLY LIVED

A significant finding during this work: the monster AI I was asked to modify was not driven by the visual logic graphs at all, but by **behaviour-tree assets referenced by a path field in the configuration table**. Edits to the graphs had no effect — the graphs were vestigial. Tracing the real execution path before changing anything saved the rest of the schedule.

> TRANSFERABLE LESSON
>
> In a decade-old live codebase, the documented system and the running system are different systems. I now verify which one I'm editing before I touch it — usually by instrumenting a probe and watching the log, not by reading the code.

BOSS ENCOUNTER: PHASE STATE MACHINE



— **Transformation is a buff, not a new entity.** Driving phase changes through the buff layer keeps threat, targeting and aggro state intact across the transition, so the encounter never resets its social contract with the players fighting it.

— **Reuse the proven AI, replace the form.** Cloning an existing, battle-tested behaviour tree and changing the presentation per phase produced a stable boss far faster than hand-authoring a new AI graph — an approach I arrived at after a hand-built version repeatedly deadlocked.

— **Fail loudly.** The debugging discipline that came out of this: a boss that gets stuck in idle is a design bug with an engineering signature, and it is found by reading timestamps, not by intuition.

Behaviour trees · Combat tuning · Boss design · Live debugging

Scaling a mode's content 20 → 110+ with an AI-assisted pipeline

The design problem was never generation speed. It was building a validation gate I could trust.

UNDER NDA Method described at architecture level only; no configuration data is shown.

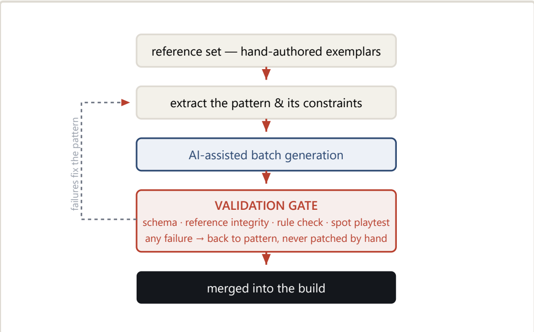
20 → 110+
CHARACTERS CONFIGURED FOR THE MODE

≈ 7×
THROUGHPUT VS. MANUAL AUTHORING

4-stage
AUTOMATED VALIDATION GATE

0
GENERATED ENTRIES SHIPPED UNREVIEWED

PIPELINE ARCHITECTURE



THE ACTUAL INSIGHT

Generation was the cheap part. The expensive part was answering **"how do I know it isn't wrong?"** at a scale where I can no longer read every entry. So the design work went into the gate: a machine-checkable definition of what "correct" means for one entry, applied to all of them.

DESIGN RULES THAT MADE IT WORK

— **Never hand-patch a generated artefact.** A one-off fix hides a systematic defect; the failure has to be pushed back into the pattern so it is fixed 110 times.

— **Encode the traps.** Every production landmine I hit (a path field that silently needs a file extension, an index that must equal its own position, a byte-order marker that crashes a loader) went into a written constraint list the pipeline checks against.

— **Keep a human at the end.** The gate rejects; only I approve. Nothing generated reached the build without review.

WHY THIS IS A DESIGN SKILL, NOT A TOOLING SKILL

Defining the validation gate **is** writing down what the design actually requires — most of what a spec leaves implicit becomes explicit the moment a machine has to check it. The pipeline was, in effect, the most rigorous design document I have ever written.

Content scaling · Validation design · Python tooling · Human-in-the-loop

HAND-SCULPTED · NO AI

Before the pipelines: work done entirely by hand

A character study sculpted, retopologised and textured by hand — the standard every generated asset in this section is judged against.

NATURE	SUBJECT	TOOLCHAIN	BUDGET
Personal study · non-commercial	Vanellope · concept "Vanilapy" by Lauren Magpie	ZBrush · Maya · Substance Painter	21,271 triangles · game-ready

Fig 1 · Final wireframe. Hand-retopologised in Maya — three unbroken edge loops around the eyes, an independent loop group around the mouth, hair as closed separate shells. Every loop exists for a deformation it will one day have to survive.

DELIBERATE TRANSLATION CHOICES

— **Brows thickened, peak raised.** The concept reads sweet; the character's core is **stubborn**. A peaked brow over a slightly lowered lid keeps that read even at rest.

— **Hair reduced to three silhouette masses** — ponytail, side falls, fringe — clean closed volumes that hold the silhouette at distance and leave room for physics later.

— **Face gets half the polygon budget.** The hoodie keeps only its structural folds; the loops are spent where the acting happens.

WHY THIS PAGE SITS HERE

Only someone who has sculpted an eye socket by hand can tell when a generated one is wrong. This piece is the calibration for every judgement made on the following pages — **the craft is not what AI replaces; it is what qualifies me to accept or reject what AI produces.**

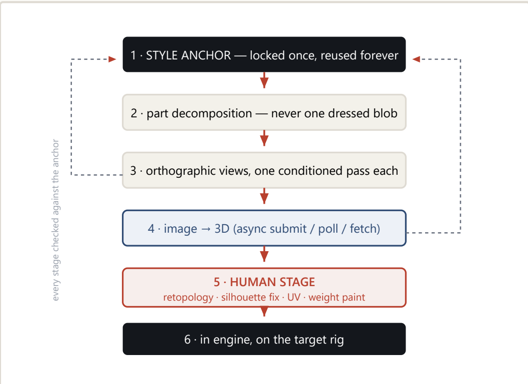
Fig 2 · Front · three-quarter · profile — the brow keeps one story from every angle. © Disney · concept: Lauren Magpie · all 3D work mine.

Producing game-ready art with generative tools

The hard problem is never getting one good asset. It is getting the hundredth one to belong to the same world as the first — and to survive rigging.

ROLE	TOOLCHAIN	OUTPUT	NOTE
Pipeline design & execution	Image generation · image-to-3D · Maya · Blender · ComfyUI	Retopologised, textured, rigged assets	Method shown; client work withheld

THE PIPELINE, AND WHERE A HUMAN HAS TO STAND



Each stage validates against the **style anchor**, so drift is caught at the stage that caused it rather than discovered at rigging. That single rule is what turns a pile of prompts into a pipeline.

LOCAL DEPLOYMENT, REAL LIMITS

ComfyUI runs locally on a 12 GB consumer GPU with custom nodes. Working inside a fixed VRAM budget forces honest engagement with batching, resolution ladders and quantisation trade-offs, instead of treating generation as an infinite cloud resource.

WHY THIS IS A DESIGN PROBLEM

Deciding **what to decompose into** is the whole job. Split a character too finely and every asset needs bespoke assembly; too coarsely and nothing is reusable. That judgement is not automatable, and it is the part I find genuinely interesting.

FOUR RULES I ARRIVED AT THE HARD WAY

— **Decompose before you generate.** A whole dressed character comes back as an un-editable blob. Hair / garment / accessory / footwear as separate passes gives you a wardrobe instead of an asset.

— **Never generate a dressed body as one mesh.** It looks right and deforms wrong — the garment inherits the body's weights and shears at the shoulder. Base body, garment shell and sleeves each need their own weights.

— **Thicken before generating, not after.** Cloth and hair edges come back as zero-thickness shells and vanish under normal-based shading.

— **Normalise scale on arrival.** Every generated part returns in its own units; making this a mandatory step rather than a remembered one removed a whole class of silent failures.

THE HONEST LIMIT

These tools reach roughly **70 %** in minutes and then stop. Silhouette correction, topology that survives deformation, and weight painting are still entirely human — and they are exactly where an asset becomes either good or unusable. **My interest is in designing that handoff, not in pretending it does not exist.**

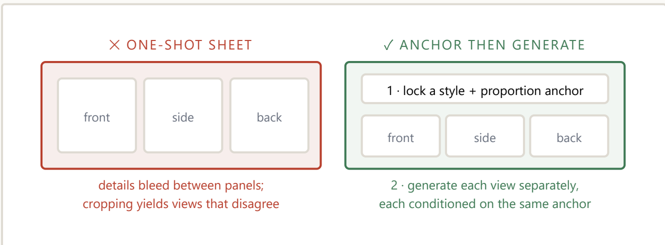
Pipeline design · Image→3D · Retopology · PBR · ComfyUI · Maya

Anchor-then-generate

Reference sheets are the input to every image-to-3D tool. Getting three views that actually agree is the whole problem — and the obvious approach is the one that fails.

Fig 1. Front, side and back, generated as three separate conditioned passes against one locked anchor. Same proportions, same palette, same costume logic — which is what a downstream 3D generator needs in order to reconcile them into a single mesh.

THE FAILURE I STARTED FROM



Asking a model for a three-view sheet in one image is the obvious approach and it fails reliably: adjacent panels contaminate each other, and once you crop them the front and side no longer describe the same character. **Anchoring style and proportion first, then generating each view as its own conditioned pass**, produces views a downstream generator can actually use.

THE END OF THE PIPELINE

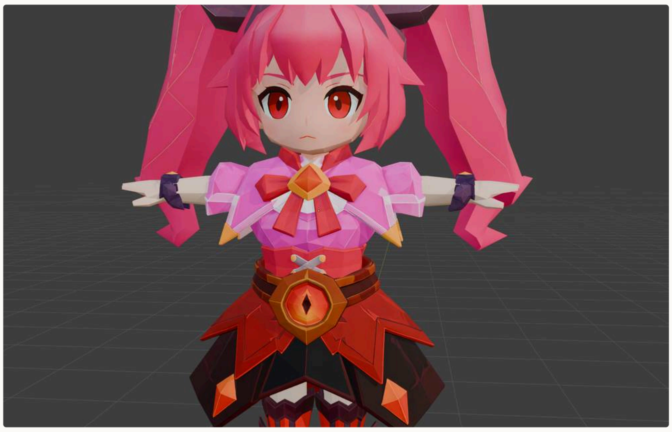


Fig 2. The finished game model these views fed into — viewport capture of the real asset: **14,452 triangles**, hand-painted texture, T-pose on the target rig.

The anchor is an asset, not a prompt. Every later character is negotiated against a file, not against the last thing the model happened to produce — consistency stops being a per-character argument.

WHY THIS MATTERS BEYOND ART

Everything difficult about generative production is a **consistency** problem, not a quality problem. Any model can produce one striking image; a production needs a hundred that belong to the same world. The method above is my answer to that, and it is the question I most want to keep working on academically.

An internal tournament & prediction platform

Designed, built, tested and operated end to end. Ran a full competitive season with a real user base.

ROLE	STACK	SCOPE	STATUS
Solo — product, design, full-stack, ops	Python / Flask · SQL · server-rendered front end	Schedule · virtual-currency prediction · ledger · admin · analytics	Shipped; ran one full season in production

106	70	6.26	0
ACTIVE USERS	SMOKE TESTS GATING DEPLOYS	AVG. ACTIVE PAGES / USER	LEDGER DISCREPANCIES

THREE DECISIONS I WOULD DEFEND

— **Money is never a float.** All amounts stored as strings and computed as decimals, behind a single wallet service that every balance change must pass through, with a signed ledger entry per movement. The system is auditable by construction rather than by discipline.

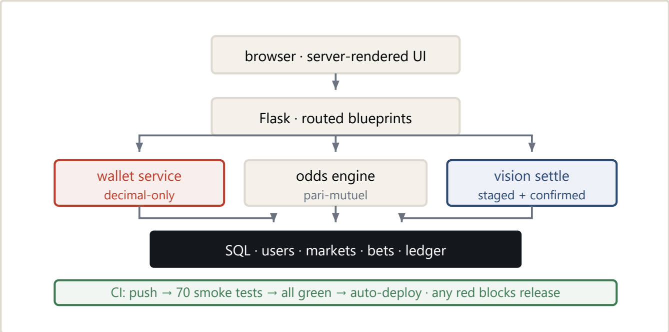
— **The deploy gate is the quality bar.** 70 smoke tests run in CI on every push; the deploy only fires if all pass. Not "run the tests" — red means it does not ship.

— **AI proposes, a human commits.** To speed up settlement I added a vision model that reads result screenshots. Its output goes to a staging buffer and a confirmation dialog — **the model never writes to the database.** Low confidence returns "unknown" and settles nothing.

DESIGN LESSON

Real users double-click. Handling that took three layers — front-end submit locking, a server-side duplicate window, and a database uniqueness constraint as the final guarantee. A design that only works when the user behaves is not a design.

ARCHITECTURE



THE ANALYTICS LOOP

I designed the event-tracking taxonomy alongside the features, so behaviour data existed from day one. It showed which pages users actually reached (averaging 6.26 active pages each) and drove the next round of navigation and UI changes. **Instrumenting at design time rather than after launch** is the single practice I would carry into any product I work on.

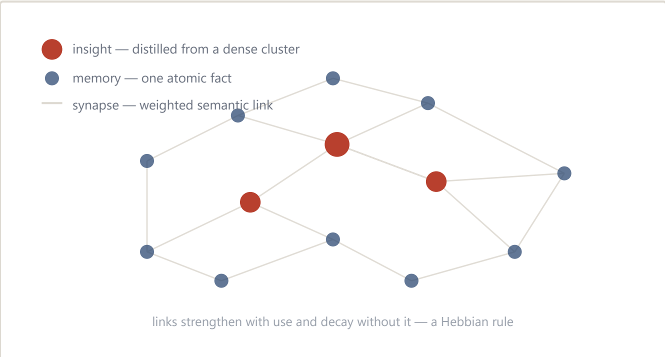
Full-stack · Systems economy · CI test gates · Analytics design · Human-in-the-loop AI

Second Brain — a self-organising knowledge system

A personal research project: a retrieval system that keeps learning, de-duplicating and correcting itself, built as a graph rather than a list.

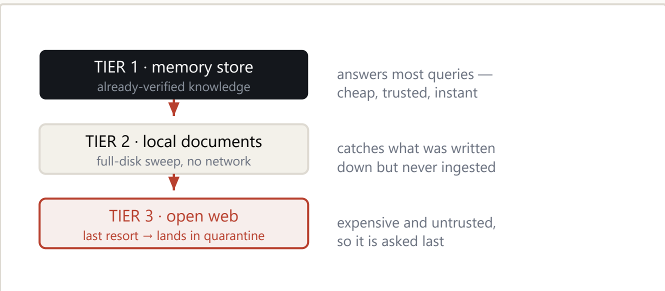
79	128	16	3-tier
ATOMIC MEMORIES	SEMANTIC CONNECTIONS	HIGHER-ORDER INSIGHTS	RETRIEVAL CASCADE

STRUCTURE



Flat vector search returns things that are similar. Building explicit connections between memories, then distilling densely-connected clusters into higher-order insights, lets the system return things that are **relevant but not obviously similar** — which is where most useful retrieval actually lives.

THE RETRIEVAL CASCADE



Escalation is **cheapest-first by design**: each tier is tried only when the one above it fails, so cost and risk are paid only when they buy something.

TWO IDEAS I AM PROUD OF

1 · A QUARANTINE BEFORE THE LIBRARY

Anything newly acquired lands in a **staging area**, not the main store. It is promoted only after passing a quality gate — length, source credibility, and near-duplicate similarity against existing memories. Most systems optimise retrieval; almost none defend ingestion. A knowledge base degrades from what you let in.

2 · A RETRIEVAL CASCADE, CHEAPEST FIRST

Queries try the existing memory store, then a local document sweep, and only then reach the open web. Most questions are answered before the third tier — which saves time, cost, and the temptation to trust a stranger's page over your own recorded experience.

WHY A DESIGNER BUILT THIS

It started as a practical problem — hundreds of scattered production notes I could no longer search — but the interesting part turned out to be an interface question: **what should a system remember on your behalf, and how does it show you that it might be wrong?** That is the research question I would most like to pursue formally.

Retrieval systems · Knowledge graphs · Data visualisation · Python / SQL

CLOSING

Disclosure, credits & contact

GENERATIVE-AI DISCLOSURE

Declared in full, in line with the programme's request.

— **Section III, benchmark page** — the Vanellope study is entirely hand-made (ZBrush, Maya, Substance Painter); no generative tools were used at any stage.

— **Section III — AI-assisted art pipelines** — generative image and image-to-3D models were used as production tools, which is the explicit subject of that work. Concept direction, part decomposition, retopology, assembly, rigging, weight painting and all in-engine work are mine.

— **Section II — content scaling** — AI-assisted generation was used to author configuration data behind a validation gate I designed. No generated entry shipped without human review.

— **Section I — game projects** — no generative AI was used in the games themselves: design, level construction, Blueprint logic, behaviour trees and animation are hand-authored. Screenshots were upscaled with AI super-resolution for print clarity; content is unchanged.

— **This document** — written by me; layout is hand-coded.

CONFIDENTIALITY

Section II describes work carried out at Tencent TiMi Studio Group and is covered by a non-disclosure agreement. **No proprietary screenshots, configuration data, source assets or unreleased content appear anywhere in this portfolio.** All diagrams in that section were re-drawn by me at a generic level of abstraction that conveys the design reasoning without disclosing the product. I would rather present my industry work honestly and incompletely than present it improperly.

CREDITS & SCOPE OF AUTHORSHIP

— Dream · 0423 · Jim's Adventure — solo projects; all design and implementation mine, with selected art and audio assets sourced or outsourced.

— Vanellope study — personal fan study; character © Disney, concept "Vanilapy" by Lauren Magpie; all 3D work mine.

— Clay Beats — team capstone; my ownership was animation production and core interaction design. The title is credited to the full team.

— Industry work (Section II) — collaborative by nature; my specific contribution is stated on each page.

— Everything in Sections III–IV is individually authored.

LINKS

Gameplay footage — all three solo titles:
flypigkun.github.io/portfolio/videos

Clay Beats **on Steam:**
store.steampowered.com/app/3492920

Portfolio online:
flypigkun.github.io/portfolio

CONTACT

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Thank you for reading. Every project here is the same question asked in a different medium: what has to be true about a production system before an idea can survive contact with it?